

- 3 (a) (i) $M = - \text{gradient} / G$ [1]
 Proper substitution of values to find gradient = 5.9×10^{15} [1]
- (ii) $GMm/r^2 = mr\omega^2$
 $r^3 = (6.67 \times 10^{-11})(8.8 \times 10^{25})(0.72 \times 24 \times 60 \times 60)^2 / (4\pi^2)$
 $r = 8.32 \times 10^7 \text{ m}$ [1]
- To move satellite to infinity, $GPE + KE + \Delta E = 0$
 $\Delta E = -GPE - KE$
 $= -(-GMm/r) - \frac{1}{2}mv^2$ [1]
 $= (6.67 \times 10^{-11})(8.8 \times 10^{25})(1200)/(8.32 \times 10^7) - \frac{1}{2}(1200)(8400^2)$
 $= 4.23 \times 10^{10} \text{ J}$ [1]
- (b) At lower orbit (smaller r), total energy decreases / becomes more negative because $E_T = -GMm/(2r)$. [1]
- Thrust must be directed opposite to the satellite's velocity, because the thrust must do negative work on the satellite to cause the required decrease in its total energy. [1]

